2 Figure 2.1 shows a Kundt's tube of length 35 cm, containing air and fine sand, having one end open at position **M** and one closed at position **N**. A loudspeaker connected to a sound generator is placed at its open end. When the sound generator and loudspeaker is turned on, four small heaps of sand eventually formed at certain positions in the tube. Adjacent heaps of sand are at distance 10 cm apart as shown in Figure 2.1.

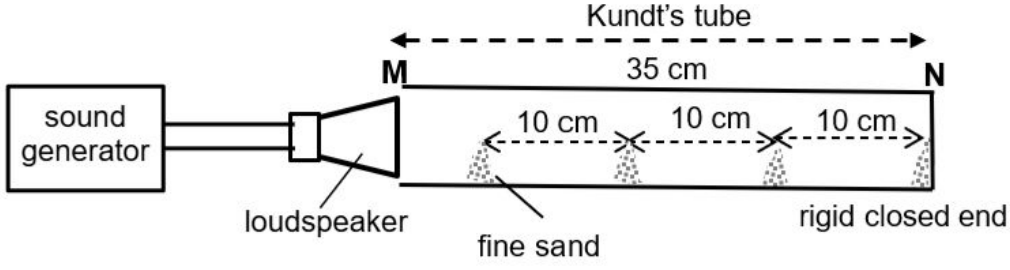


Figure 2.1

(a) Explain how a stationary wave is formed in the tube.

.....

.....

.....

.....

.....

.....

..... [2]

L1 **The incident sound wave is reflected at the closed end.** **B1**
Since the incident and reflected waves are of the same nature, same frequency, same speed and approximately equal amplitude and are travelling in opposite directions along the same line (OR overlap in opposite directions), a stationary wave can form.

A stationary wave will be formed if the tube length equals to an odd number multiple of a quarter wavelength. **B1**
(as long as proper boundary conditions are stated)

(b) Determine the wavelength of the sound produced by the loudspeaker.

wavelength = ... cm [1]

L1 The heaps of sand form at the positions of displacement nodes.
Distance between successive displacement nodes = $\frac{\lambda}{2}$

$\frac{\lambda}{2} = 10\ cm \implies \lambda = 2 \times 10 = 20\ cm$ **A1**

(c) (i) The sound wave produced by the loudspeaker in Figure 2.1 has amplitude of A_o.

On Figure 2.2, sketch a graph of the variation of amplitude, in terms of A_o, of the stationary sound wave with distance x along the Kundt's tube, starting from point **M** of the Kundt's tube (x = 0 cm), until its closed end at point **N** (x = 35 cm). You may assume there is no energy loss by the sound wave as it is reflected at the closed end of the tube.

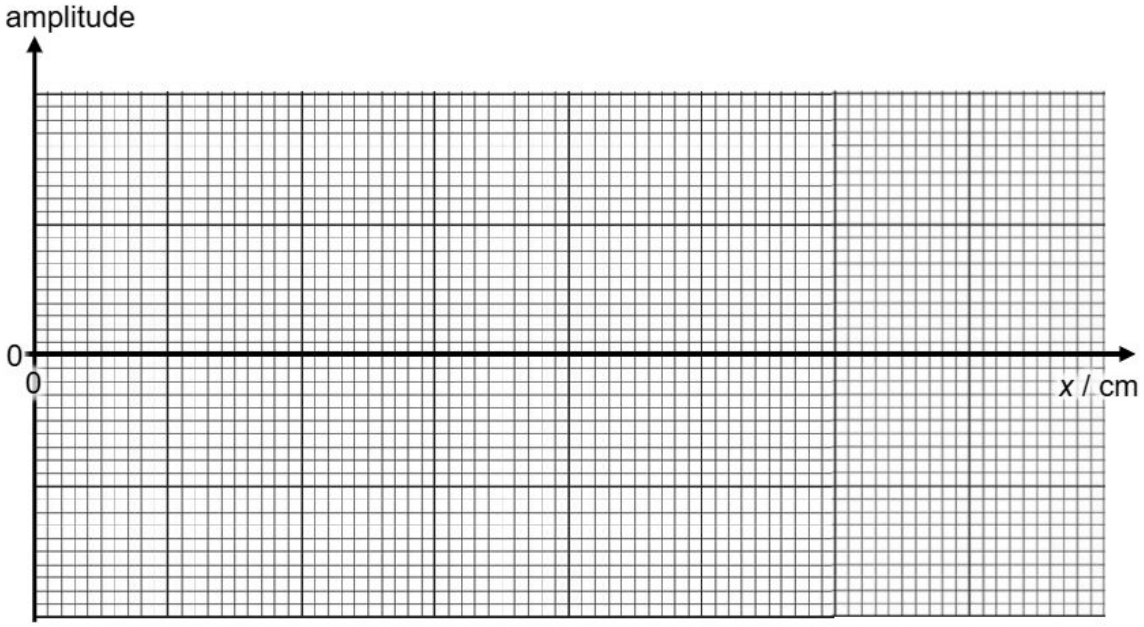
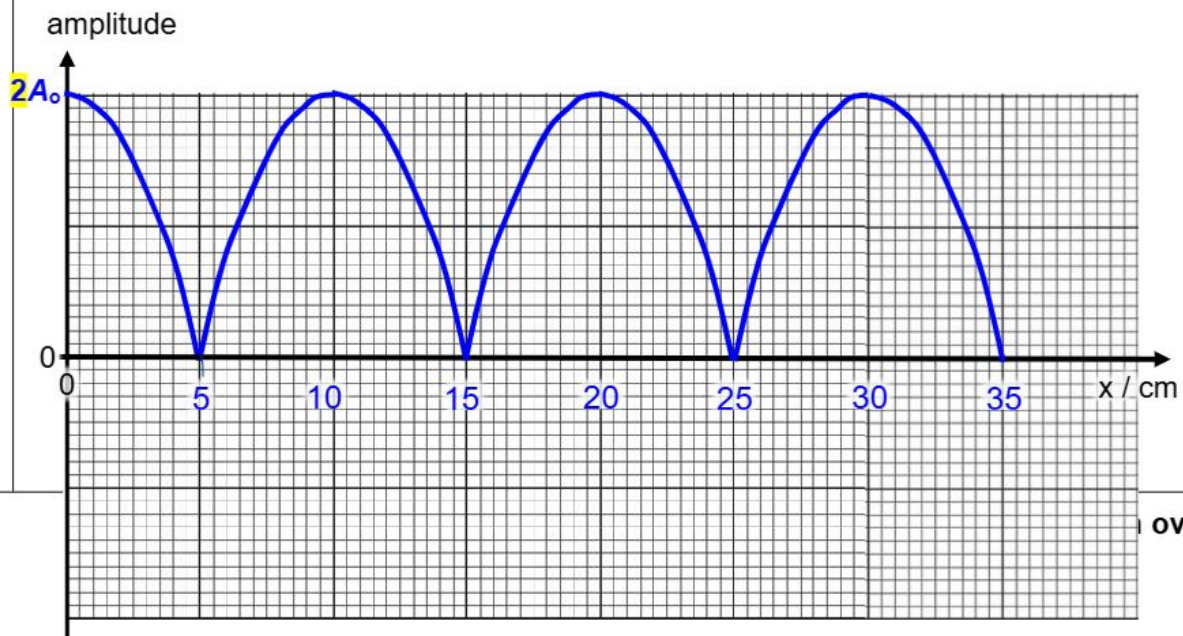


Figure 2.2 [3]

Solution:



over

	L2	**B1- Correct shape (no negative; <u>modulus cosine</u> shape) AND 4 maximum amplitudes.** ***[This mark is for shape, correct positions of zero & maximum amplitudes is marked for in 3rd B1.]*** ***<u>Not</u>*** *cosine-**squared** shape, unless it is Intensity variation graph then 'squared' since Intensity proportional to Amplitude-squared.*		
	L3	**B1- Labelled correct amplitude of the resultant stationary wave (= $2A_o$).**		
	L2	**B1- Labelled correct positions of zero AND maximum amplitude on the x-axis. Good scale.**		
		(ii)	By describing the movement of air molecules in the stationary sound wave, state and explain where the air pressure varies the most.	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	[3]
		L2	At **<u>displacement</u> <u>nodes</u>** of the stationary wave. **Candidates must distinguish between displacement node & pressure node.*	**B1**
			<u>Air molecules</u> at <u>either side</u> of <u>each displacement node</u> are <u>always vibrating antiphase.</u>	**B1**
			Hence, the **displacement nodes are <u>the positions of maximum compression (OR where there is highest pressure), *as well as*, the positions of maximum rarefaction (OR where there is lowest pressure).</u>** Thus, the greatest variations in pressure occur at the displacement nodes.	**B1**

(iii) The maximum air pressure in the Kundt's tube is P_o.

On Figure 2.3, sketch a graph of air pressure variation, in terms of P_o, against distance x along the Kundt's tube from point **M** (x = 0) to point **N** (x = 35 cm).

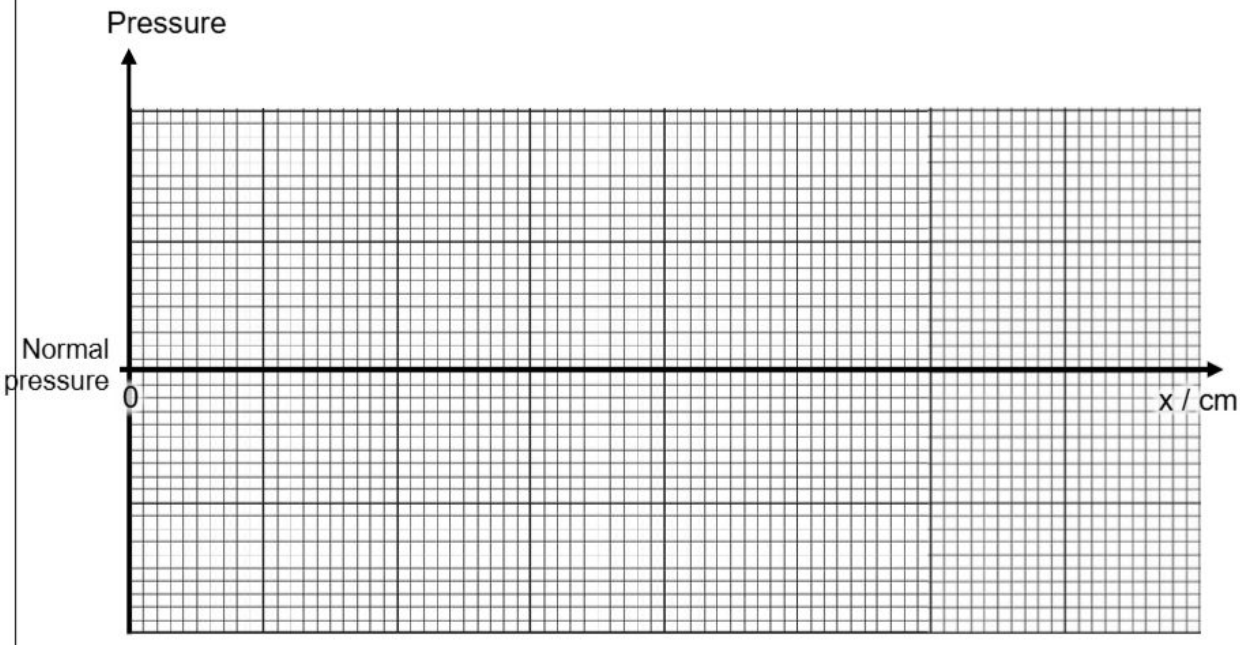
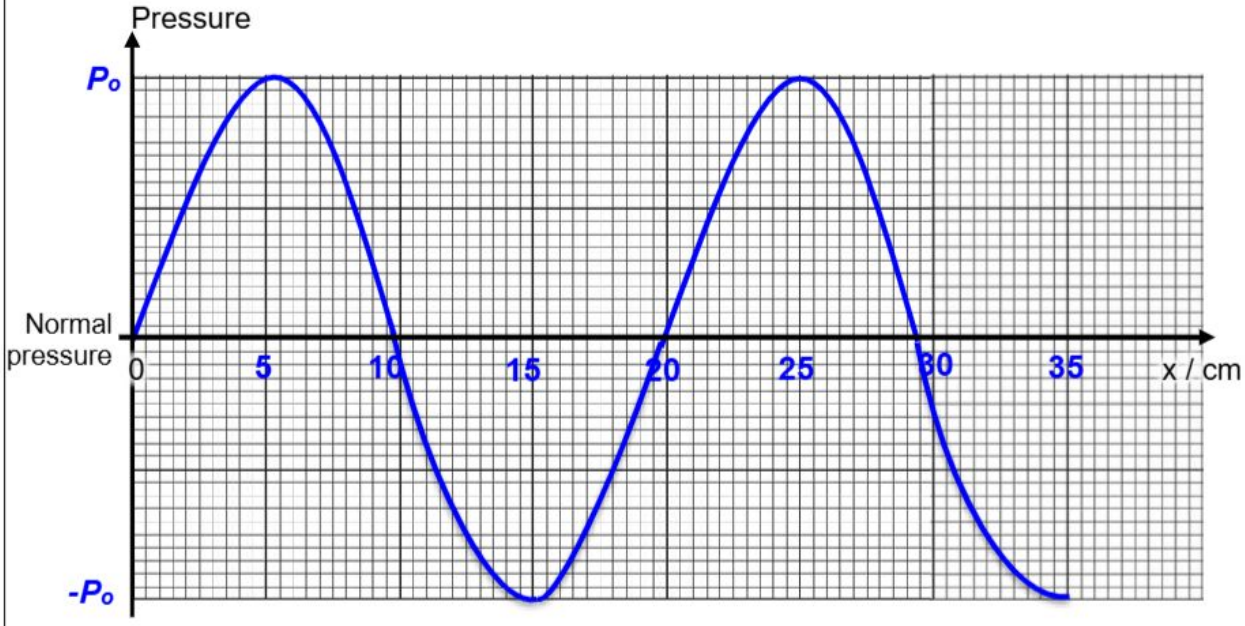


Figure 2.3 [2]

L3 Solution:



B1: Correct shape (sinusoidal; *either* positive sine *or* negative sine curve) with positive and negative P_o.

	B1: Labelled correct positions of Zero AND Maximum Pressure variations along the x-axis. ***Allow ECF from (c)(i).***

[Total: 11]